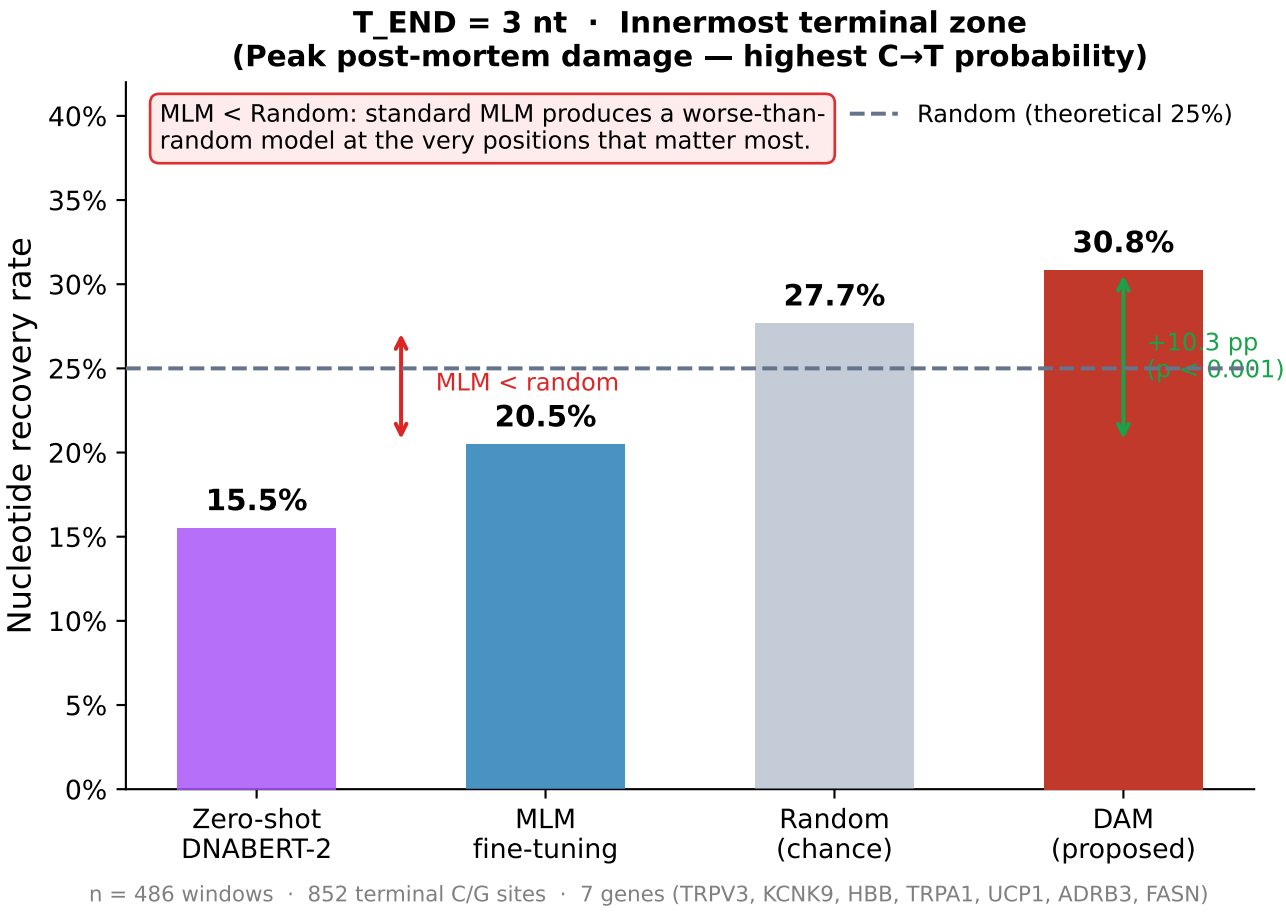
T_END = 3 nt · Innermost terminal zone
(Peak post-mortem damage — highest C→T probability)
MLM < Random: standard MLM produces a worse-than-random model at the very positions that matter most.
Random (theoretical 25%)
Nucleotide recovery rate
40%
35%
30%
25%
20%
15%
10%
5%
0%
15.5%
20.5%
27.7%
30.8%
MLM < random
+10.3 pp
(p < 0.001)
Zero-shot DNABERT-2
MLM fine-tuning
Random (chance)
DAM (proposed)
n = 486 windows · 852 terminal C/G sites · 7 genes (TRPV3, KCNK9, HBB, TRPA1, UCP1, ADRB3, FASN)